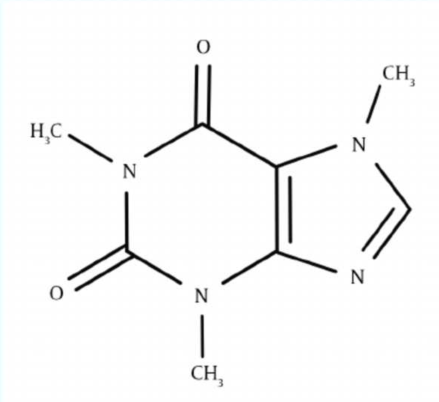
Evaluating the Impact of Chronic Juvenile Caffeine Exposure on Learning and Memory using RT-qPCR analysis of CREB1 and MAO Gene Expression in *Lymnaea stagnalis*

Background



Caffeine ($C_8H_{10}N_4O_2$) is a **psychoactive substance**, which indicates that it affects the **central nervous system (CNS)**. It can cause changes in **mood, awareness, and behavior** (Temple, 2010).

Recently, caffeinated products have **grown in popularity among juveniles** (Temple, 2010). The **FDA has yet to recommend a caffeine consumption limit for youth populations**, despite having multiple recommendations set for adults, contributing to concern about **the impacts of caffeine on neurodevelopment** (Johns Hopkins Medicine, n.d.).

Recent research indicates that **adolescent caffeine impairs the development of the cerebral cortex (Rosania, 2013).**

The cerebral cortex is primarily responsible for **thinking, learning, and memory**, suggesting that cognitive challenges may occur if developmented is inhibited in this region.

***Lymnaea stagnalis*, a popular model organism used in behavioral and developmental research**, may be able to further indicate the effects of caffeine consumption due **to its high quality reference genome.**

***Lymnaea stagnalis*, commonly known as "The Great Pond Snail"**

RT-qPCR can be used to further analyze **gene expression** in *Lymnaea stagnalis*.

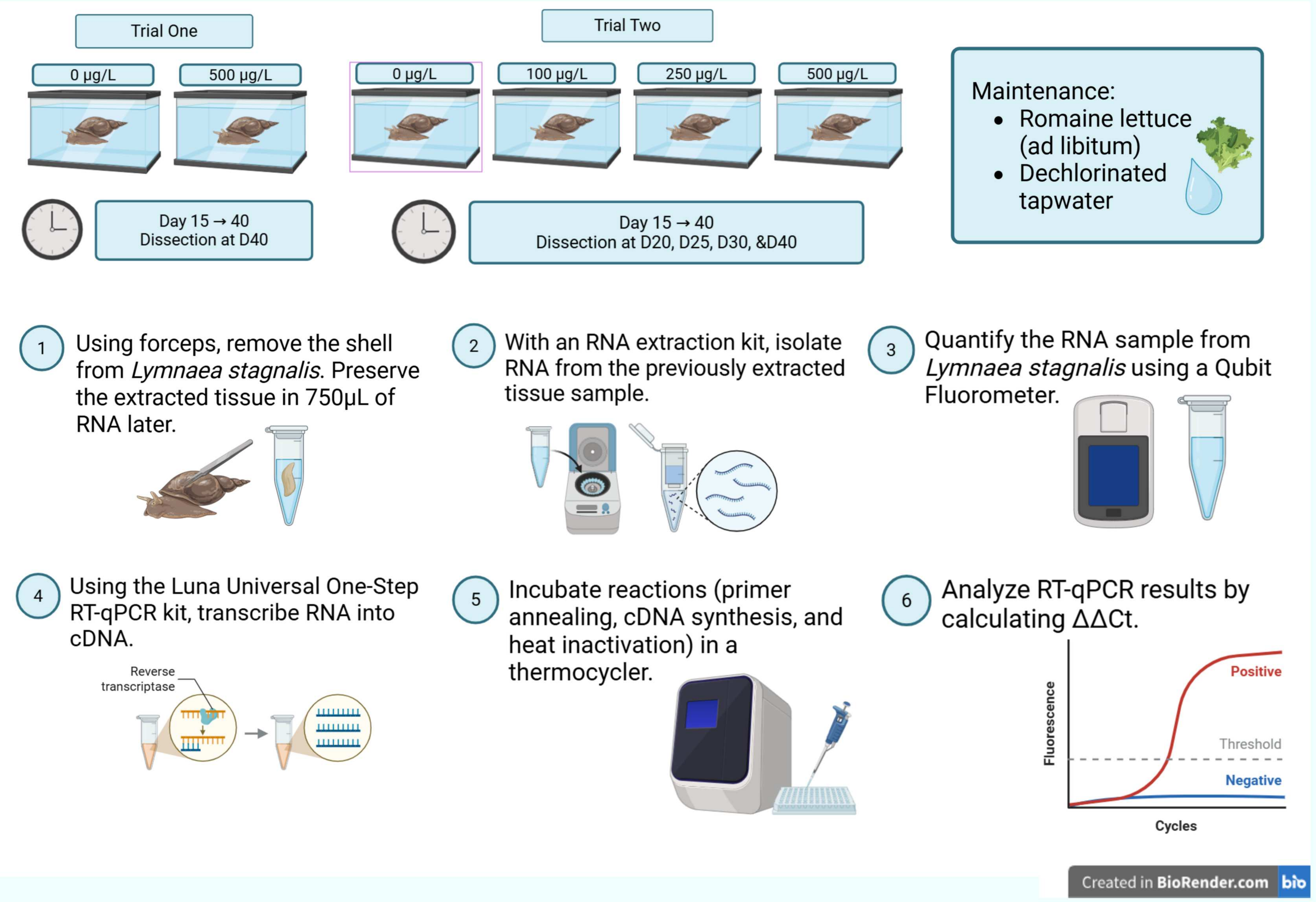
Reference genes:
CREB1 → cAMP response element binding protein, **crucial to LTM consolidation.**
MAO → monoamine oxidase, **regulates neurotransmitter degradation** (Chikamoto et. al 2013).

Hypothesis

If ***Lymnaea stagnalis*** are exposed to caffeine from **day 15 to day 40** of their life cycle, **CREB1 will be downregulated** and **MAO** will be **upregulated**. This is due to the fact that chronic exposure to caffeine during juvenile development can **overstimulate neurons** by **blocking adenosine receptors, which normally help regulate neuronal activity** (Temple, 2010). Prolonged exposure to caffeine can trigger a **negative feedback mechanism in the nervous system,** leading neurons to **downregulate CREB1 and upregulate MAO in order to prevent overactivation**.

Independent Variable: Caffeine concentration
Dependent Variable: Gene expression as measured using RT-qPCR

Experimental Design



Data and Results

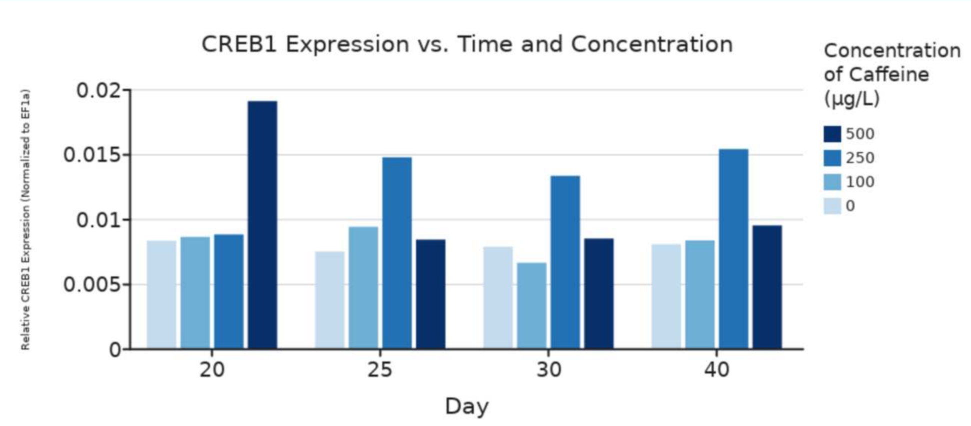


Figure 1. CREB1 Expression Normalized to Ef1a vs. Day and Concentration. "Day" represents the time of sample dissection. "Concentration" represents the amount of caffeine administered to each group of Lymnaea stagnalis.

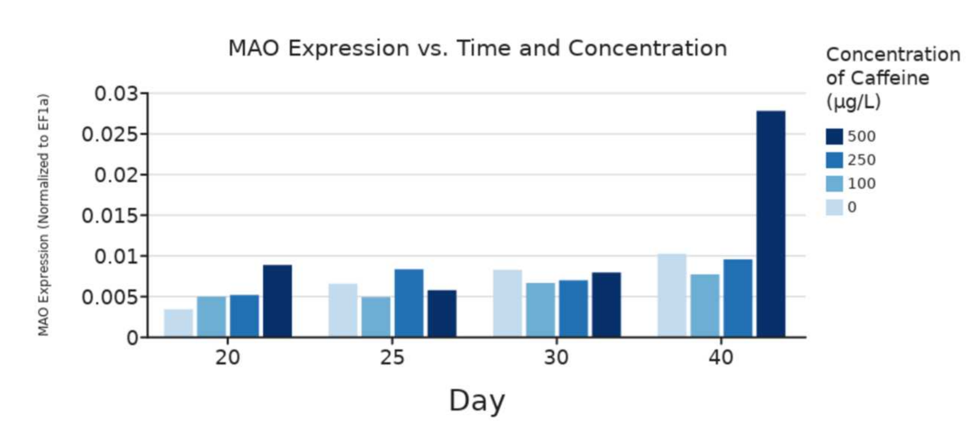


Figure 2. MAO Expression Normalized to Ef1a vs. Day and Concentration. "Day" represents the time of sample dissection. "Concentration" represents the amount of caffeine administered to each group of Lymnaea stagnalis.

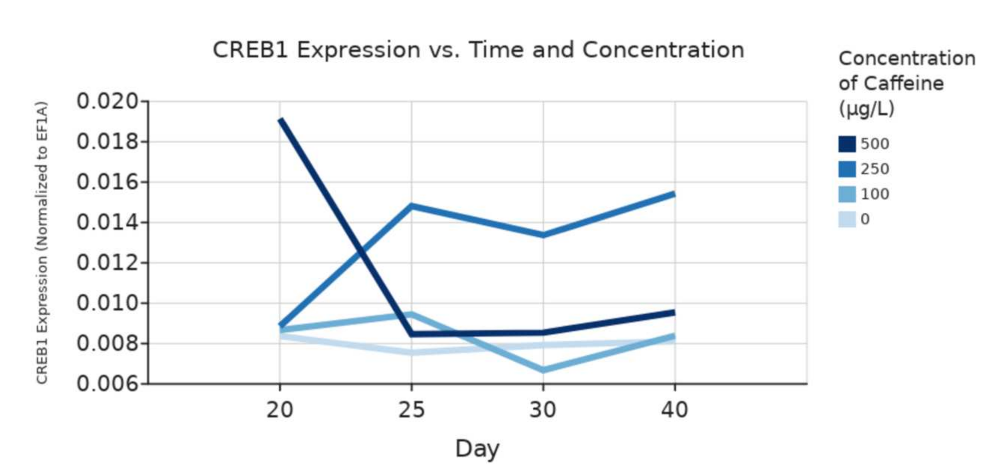


Figure 3. CREB1 Expression Normalized to Ef1a vs. Day and Concentration. Lines represent change in gene expression over time for each concentration of caffeine.

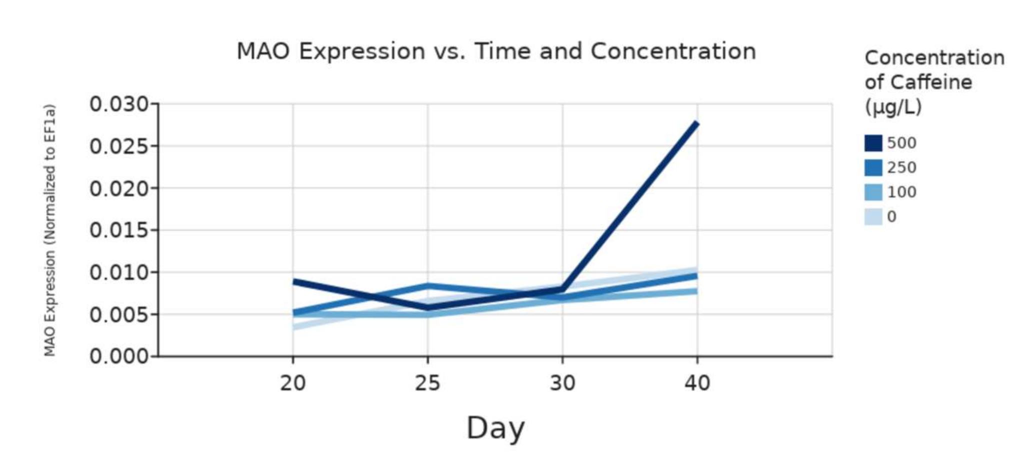


Figure 4. MAO Expression Normalized to Ef1a vs. Day and Concentration. Lines represent change in gene expression over time for each concentration of caffeine.

Conclusions

Chronic caffeine exposure during juvenile development was associated with increased CREB1 and MAO expression relative to the housekeeping gene EF1a.

Key Findings

CREB1
- Higher caffeine concentrations were associated with **increased CREB1** expression, particularly during **earlier developmental stages**
  - Peak expression occured in the 250 µg/L group

MAO
- Higher caffeine concentrations were associated with **increased MAO** expression **at day 20 and day 40**
  - Peak expression occurred in the 500 µg/L group at day 40

Interpretation

CREB1
- **Contrary to the original hypothesis**, elevated caffeine exposure was associated with increased CREB1 expression
  - This may suggest enhanced transcriptional activity in memory-related neural pathways

MAO
- **In accordance with the original hypothesis**, elevated caffeine exposure was associated with increased MAO expression
  - This may suggest a compensatory response to prolonged neural stimulation

Limitations

- Limited biological replicates prevented statistical analysis from being performed, so interpretations were based on expression trends
- The number of RT-qPCR reactions performed was limited by time and cost constraints

Future Work

- Increase sample size and RT-qPCR replicates to improve statistical reliability
- Examine the underlying feedback mechanisms behind caffeine exposure to better understand its implications and direct influence on CREB1 and MAO expression
- Investigate whether long-term caffeine exposure produces lasting behavioral effects in *Lymnaea stagnalis* through additional assays

References

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